

AI for Bharat Hackathon

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Team Name : Mudda

Team Leader Name : Shivesh Kumar Dhiman

Problem Statement : Mudda is a **social-media** inspired platform that empowers citizens to **raise and track civic issues transparently**. It bridges the gap between the public and government authorities by ensuring **accountability, real-time updates, and community engagement**.



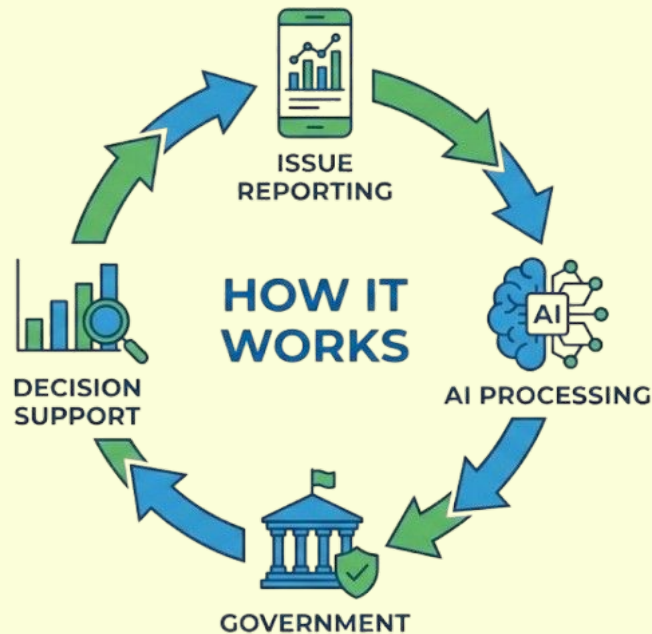
Mudda: AI-Powered Civic Engagement

MUDDA

- The Goal: Digitally transform how citizens report, track, and resolve public issues.
- The Vibe: Transparent, community-driven governance with the simplicity of social media.
- The Solution: Bridges the gap between citizens and government via:
 - Real-time issue tracking
 - AI-powered management
 - Data-driven decision support

[HTTPS://MUDDA.DEV:](https://mudda.dev)

India's First Autonomous AI Governance Co-Pilot



Why Mudda Works: Differentiation & Impact

Innovation & Differentiation

- **Social-Media Inspired:** Citizen-centric design with public visibility and community validation (upvotes/comments).
- **Smart Categorization:** AI-powered duplicate detection and issue sorting.
- **Clear Accountability:** Real-time tracking with direct authority mapping.
- **Actionable Dashboards:** Data-driven interfaces for governance.

Core Solution

- **Centralized Hub:** One platform for all civic issues.
- **Verified Proof:** Geo-tagged reporting with photo/video evidence.
- **Intelligent Routing:** AI-based severity prioritization and dispatch.
- **Total Transparency:** Automated status updates and notifications.
- **Smart Allocation:** Turns raw data into structured resource planning.

Our USP

Mudda = AI-Powered Civic Intelligence Engine

- We transform chaotic, unstructured citizen complaints into **actionable, transparent, and scalable** governance insights specifically designed for Bharat.

Feature Ecosystem: Bridging Citizens & Governance

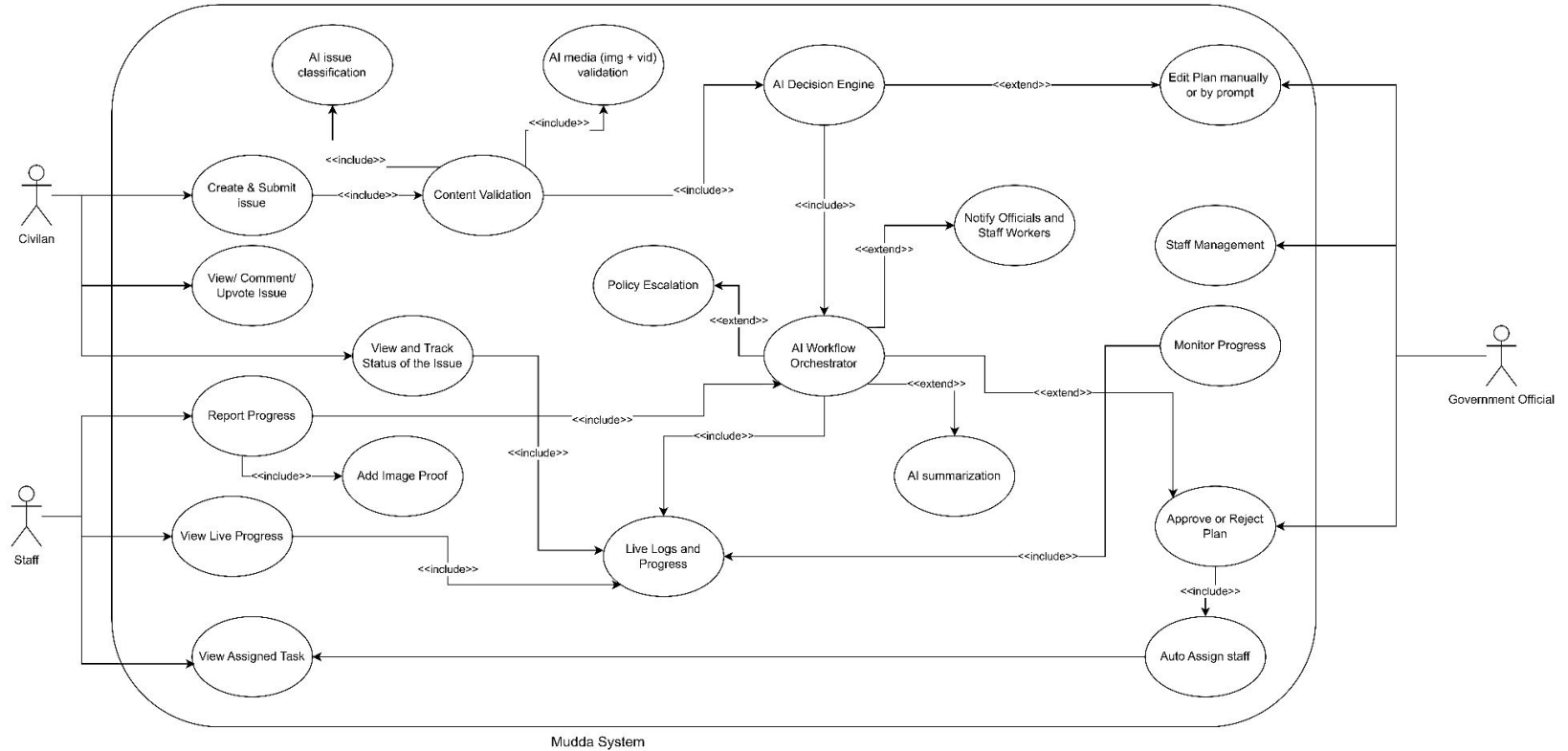
The Citizen App (User Experience)

-  **Precision Reporting:** Geo-tagged issues with photo/video evidence.
-  **Community Validation:** Upvote and comment system to highlight urgent issues.
-  **Real-Time Updates:** Transparent status tracking and resolution alerts.
-  **Direct Feedback:** Closed-loop communication with authorities.

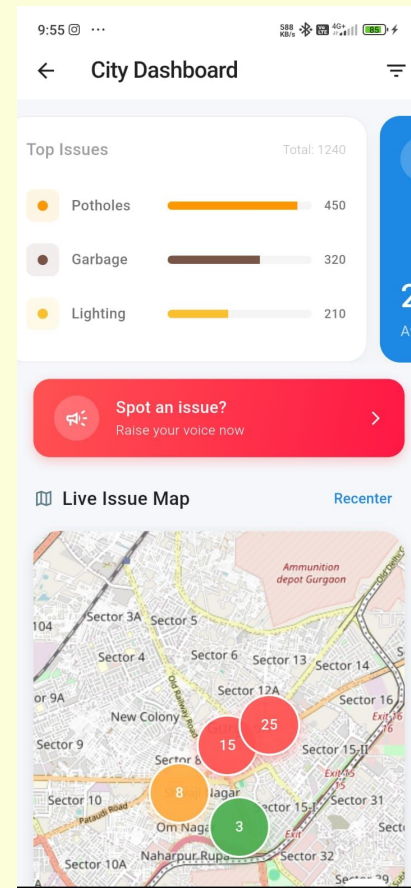
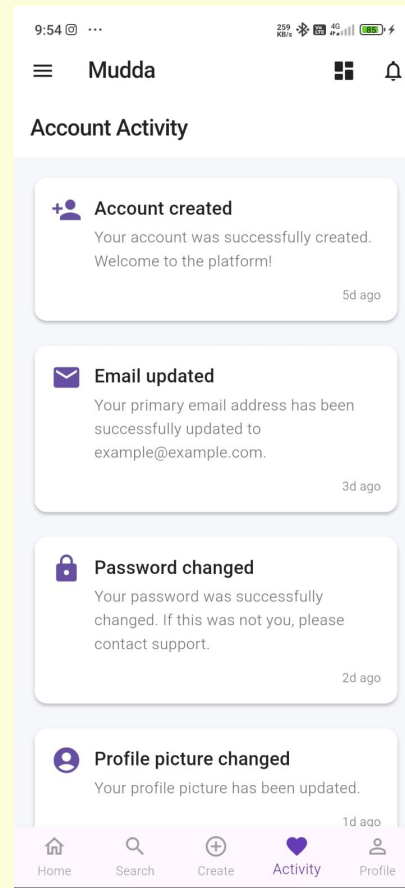
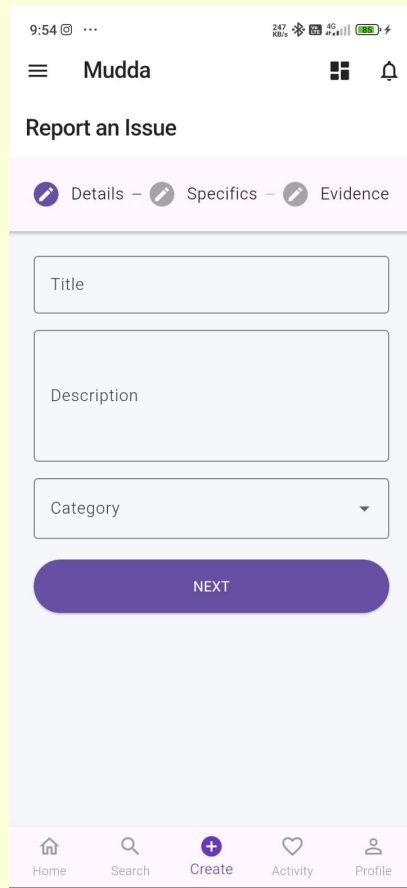
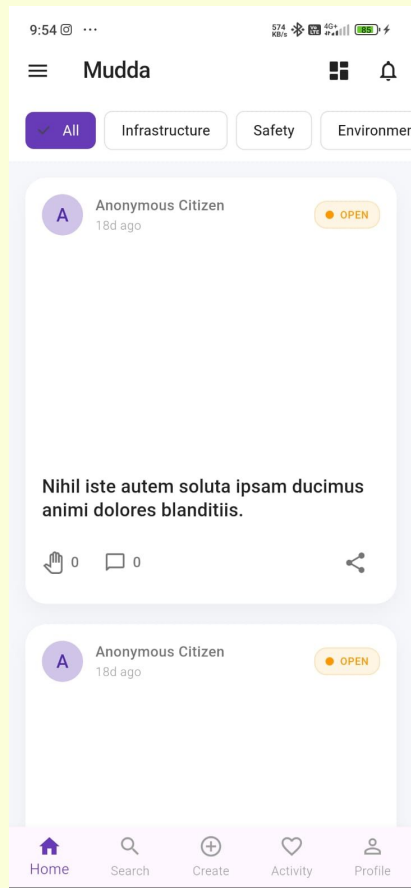
The Authority Portal (Operational Efficiency)

-  **AI Triage:** Auto-categorization and smart duplicate detection.
-  **Intelligent Routing:** Severity scoring sends issues to the right department.
-  **Analytics Dashboard:** Heatmaps and trend analysis for resource allocation.
-  **Authority Mapping:** Clear ownership and accountability assignment.

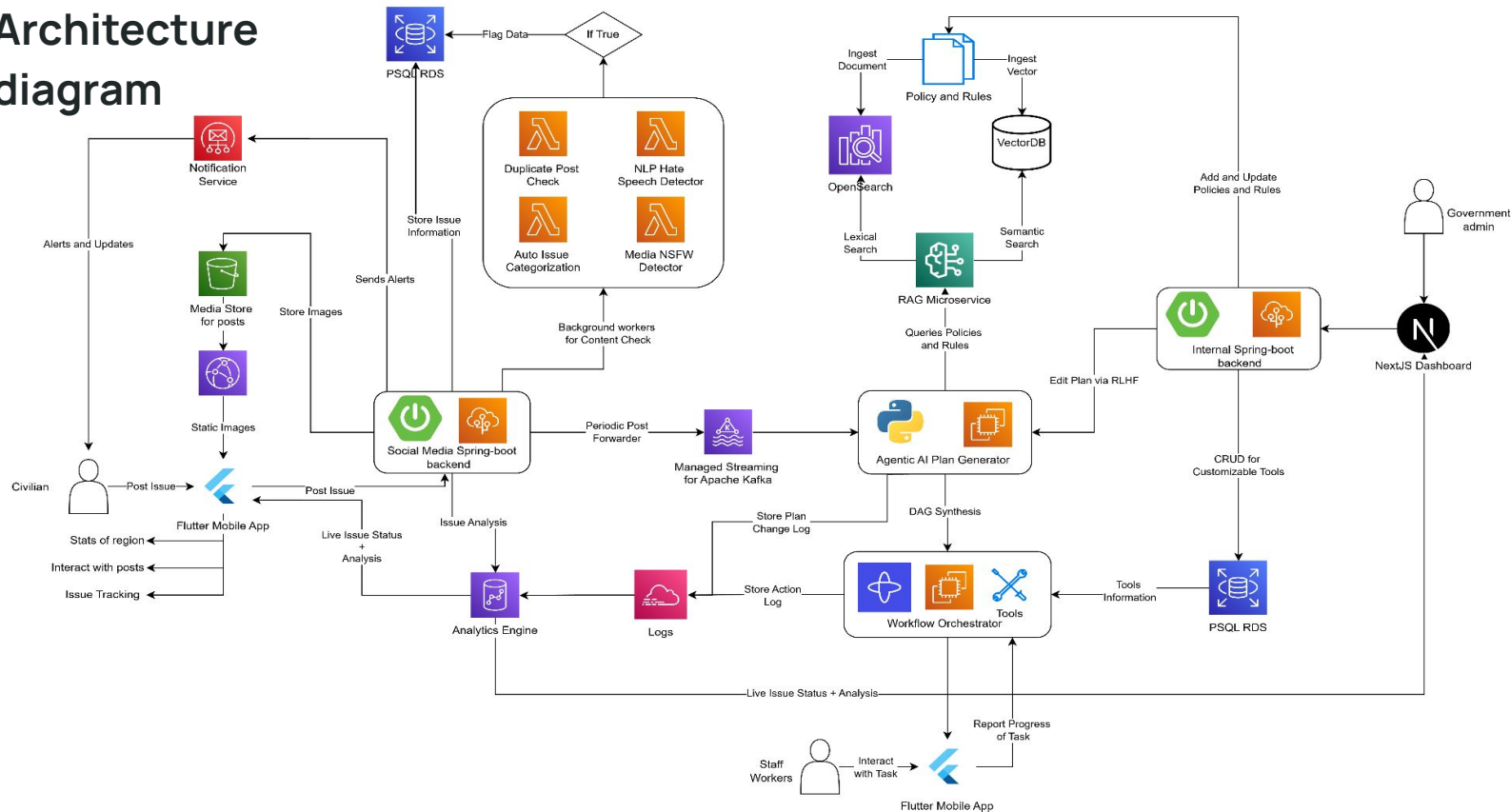
Use-case diagram



Wireframes/Mock diagrams of the proposed solution



Architecture diagram



Technologies to be used in the solution:

Frontend	Flutter – Cross-platform mobile app (Android & iOS)
Backend	Spring Boot – Scalable REST APIs & microservices architecture
Database	PostgreSQL + PostGIS – Structured & geo-spatial data
AI Layer	Pre-trained NLP models, Custom Agentic AI workflows for: 1. Automatic issue categorization, 2. Duplicate detection, 3. Smart routing & prioritization
Cloud & Infrastructure	AWS Elastic Beanstalk, AWS S3, AWS SES, AWS ACM, AWS CloudFront

Estimated implementation cost (optional): ₹30,000 – ₹45,000 per month

Layer	Services Included	Est. Cost (INR / Month)
Frontend & CDN	S3 + CloudFront	₹4,000 – ₹5,500
API & Compute	API Gateway + Elastic Beanstalk + Lambda	₹10,000 – ₹14,000
Database & Cache	RDS (Single-AZ) + ElastiCache	₹10,000 – ₹13,000
Event & Workflow	MSK + EventBridge + Step Functions	₹3,000 – ₹4,500
AI Layer	Amazon Bedrock (Optimized Usage)	₹8,000 – ₹15,000
Monitoring & Security	CloudWatch + WAF + IAM/KMS	₹3,000 – ₹4,000
Communication	SES + SNS	₹1,000 – ₹2,000



Resolve_Water_Leakage_and_Overfilling

Workflow to address and resolve a reported water leakage and overfilling issue at a specific sector by dispatching external plumbing services, securing necessary approvals, and notifying stakeholders.

EDITOR CANVAS



Reset

Save
Workflow

Components

Search API components...

REST

Fetch Issue Details

Retrieve details of an issue including description, reporter, and location.

REST

Plumber Service Contact

API component for contacting plumber services to report plumbing issues and

REST

Approval Service - Human Review

Human review and approval step for workflow decisions. This component

REST

External Service - Plumber Service

Contact external plumber service for emergency repairs and maintenance

React Flow

Innovation partner **I12S**
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Thank You

